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RESEARCH-ARTICLE

Progression Balancing × Baldur's Gate 3: Insights, Terms and Tools for Multi-Dimensional Video Game Balance

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Open Access Support provided by:
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Published: 26 April 2025

[Citation in BibTeX format](#)

CHI 2025: CHI Conference on Human Factors in Computing Systems
April 26 - May 1, 2025
Yokohama, Japan

Conference Sponsors:
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Progression Balancing × Baldur’s Gate 3: Insights, Terms and Tools for Multi-Dimensional Video Game Balance

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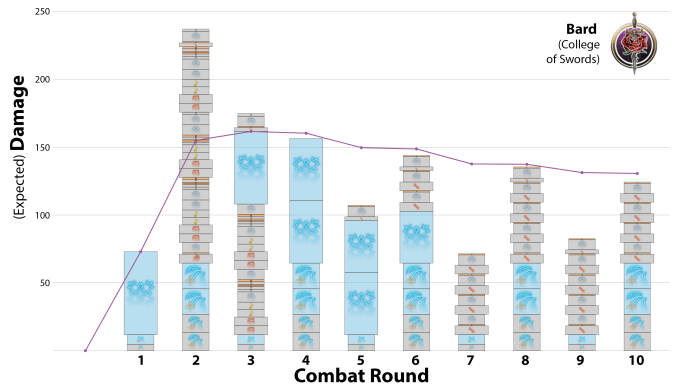


Figure 1: Keeping game elements (e.g. *classes*) balanced is a challenge that grows with the complexity of modern games. One shortcoming of traditional balancing is the comparison between elements reduced to a *single* dimension (e.g. win rate, popularity, or damage output in one benchmark condition). Using the combat logic of *Baldur’s Gate 3* (left), we built an AI-optimized simulator (right) that can calculate and visualize the most efficient behavior of a class per fight, and subsequently explore the multi-dimensional *progression* throughout the class levels, build options, opponent encounters, and more.

Abstract

Internal game balancing is one of the major components that affect player experience, as it is responsible for a large share of development time, the majority of game update patches and long-term player satisfaction. This makes tools and methodologies of assessing and advancing game balance a valuable endeavor for industry and academia. During the past decades, scientific research produced numerous outputs to inform and enhance game balancing, yet most of them only adhere to a single dimension of balance: fixed (end-game) scenarios. However, games are usually experienced throughout a continuous spectrum of ever-changing constellations, which should be reflected. Using simulation, game-playing AI, visual analytics and informative metrics, we introduce a methodology and implementation of *Progression Balancing*, incorporating multi-dimensional game aspects. For the sake of exposition and ecological validity, we applied it in one of the most successful recent games (*Baldur’s Gate 3*), and evaluated its efficacy with help of its player community.

CCS Concepts

• **Applied computing** → *Computer games*; • **Computing methodologies** → *Visual analytics*.

Keywords

Video Games, Balancing, Artificial Intelligence, Analytics, Community Survey

ACM Reference Format:

Johannes Pfau. 2025. Progression Balancing × Baldur’s Gate 3: Insights, Terms and Tools for Multi-Dimensional Video Game Balance. In *CHI Conference on Human Factors in Computing Systems (CHI ’25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 12 pages. <https://doi.org/10.1145/3706598.3713162>

1 Introduction

Acquiring a well-rounded understanding about how game mechanics influence the balance between choosable game elements is a common long-term industrial endeavor before and after a game’s launch. Yet, it remains highly important to ensure a wide range of possible options, meaningful decisions and satisfying play styles, encouraging exploration and validating preferences. With tendentially growing complexity and an expectation of having a broad range to select from, assessing and maintaining this perception of balance constitutes a hard problem, so far that it became one of the major reasons for update patches of games of most popular

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CHI ’25, Yokohama, Japan

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ACM ISBN 979-8-4007-1394-1/25/04

<https://doi.org/10.1145/3706598.3713162>

genres¹²³. “Choosable game elements” is formulated deliberately vague after all, as it can take the form of playable Champions such as in the Multiplayer Online Battle Arena (MOBA) *League of Legends* [22], equippable weapons in the First-Person Shooter (FPS) *Counter-Strike: 2* [67] or active and passive changes to classes, abilities and skills of the Massively Multiplayer Online Role-Playing Game (MMORPG) *World of Warcraft* [8]. From an industrial perspective, this has been described by a series of Game Developer’s Conference (GDC) talks, blogposts, books and videos. In *The Art of Game Design*, Jesse Schell lists twelve axes of balance with different goals and different methods to realize [60]. Likewise, in *Game Balancing*, Schreiber and Romero [62] discuss different kinds of balancing (as methodologies), dials (as quality descriptors), and modifiers as goal dimensions to be tuned towards. Griesemer adds an understanding of balance between “too predictable” (dominant choices) and “too random” (irrelevant choices), and puts more focus on (six types of) players and the different ways of getting balancing feedback from them [23]. In his dissertation about understanding game balancing, Jaffe sums up most of the former and adds further perspectives, to approximate and quantify “a meaningful diversity of gameplay experiences” [31]. Yet, understandings of what is well-balanced and how to even assess or quantify balance in a standardized fashion is a controversial and discordant topic according to Becker and Görlich [7], who contrast the incongruous voices of designers, developers, practitioners and researchers.

When discussing the importance of balance in published work, it is often tailored to competitive online games, as the confrontation with “unfairly” balanced opponents is a major cause for frustration [14]. Yet, it holds for any video game in general, as the capabilities of a character (or unit, or build), should be viable enough to keep players engaged in a state of flow – which can be harmed if they are too powerful (due to boredom) or too weak to master the game’s challenges (causing frustration) [13, 28]. Yet, while a perfect state of balance seems impossibly hard to attain – it often is not only unimportant, but even undesirable – as small power gaps can manifest as a “critical part of game balancing” [21], especially if all alternatives have their “place and time to shine”. This “place and time” could be different challenges that a game offers, it could be transitive advantages between game elements, it could be different seasons of a game’s lifetime, or it could evolve over the *progression* of a whole game. As this enumeration of possibilities that might allow for the population of many diverse but balanced game elements was certainly not exhaustive, this paper sets out to collect and understand dimensions of game balancing, and how to make use of them.

More specifically, we pursue to answer the following research questions:

RQ1: Which dimensions of in-game concepts are highly influential for a game’s notion of balance?

RQ2: Which means are appropriate to advance the understanding

of balance variables?

RQ3: Which features of visual analytics tools are necessary to explore many dimensions of game element balance at once?

In order to collect ecologically valid and quantifiable insights towards these questions, we conducted a case study by reaching out to the community of a very recent, well-received, popular and highly complex game (*Baldur’s Gate 3* [37]). Based on an initial benchmark calculated through extensive combat simulations, we aggregated vital further dimensions from the player community, incorporated these into multi-dimensional balance benchmarks, and provide insights, metrics and visualizations. While this frames the following endeavor into a specific game (or at least the larger RPG genre), our overall approach is kept generic enough to be extended to other genres as well, as discussed later. We contribute to the fields of games user research by broadening our understanding of the particulars and notions of balance, and to the field of (visual) game analytics by advancing tools to compute, illustrate and interpret higher-dimensional balance constructs. With higher-dimensional, we refer to the combinatorial interaction between multiple game elements and mechanics (e.g. classes, levels, items, combat scenarios, ...) that can all *progress* at different paces and in different directions. Thus, we coin a set of values of interest as one *balance dimension*, the combination of multiple dimension as *higher-dimensional*, one point in this higher-dimensional space as a specific *constellation*, and the overall investigation of the balance of a game element across multiple dimensions as *progression balancing*.

2 Related Work

2.1 (AI-assisted) Balancing

When it comes to academic investigations on balance, a majority of work focuses on the in-situ balancing (or adjustment) of challenge factors [1], either driven by in-game variables [4, 54, 68], affective states [35], or biofeedback [73]. However, this field of dynamic *difficulty* balancing (or adjustment) is only halfway related to the common industrial problem of balancing game *elements* to keep them relevant, viable and attractive [60, 61]. Yet, a series of academic approaches did address the balancing of viable game options, such as Monte-Carlo balancing of a (fixed) test-bed tower defense game from Beau and Bakkes [6], the optimization of a custom card game by Jaffe et al. [30] or the approach of Leigh et al., who iteratively reduced dominant strategies in a simple arcade setup [38]. While these took place in environments artificially constructed for the evaluation (and therefore fall short in ecological validity), other works did approach data and balance assessments from actually published games.

To facilitate a data-driven understanding about the state of balance, Argue deployed different machine learning approaches in *Natural Selection 2* [5], Peabody utilized DBSCAN clustering to identify shifting trends in the *League of Legends* metagame balance [22, 44], and Dockhorn et al. represent overrepresented strategies or constellations in *Hearthstone* through fuzzy multisets [9, 15]. Egliston adds an investigation with a special focus on the metagame of *WoW* [8, 19], and describes observed cases of (im-)balanced strategies. In their work on maintaining long-term commitment to games-as-a-service, Ranandeh and Mirza-Babaei deploy AI agents to derive the

¹Almost every patch of *League of Legends* (MOBA) is driven by Champion balance: <https://www.leagueoflegends.com/en-us/news/tags/patch-notes/>

²The majority of all *Guild Wars 2* (MMORPG) updates include balance tweaks: https://wiki.guildwars2.com/wiki/Category:Skill_and_trait_changes

³*Overwatch* (FPS) pushes bi-weekly balance patches with only minor bug fixes: <https://overwatch.blizzard.com/en-us/news/patch-notes/>

best balance between elements of in-game economies (currency sources, inventories, converters, currency sinks, and traders) [55]. While they do not focus on the outcome of specific choosable game elements, they still emphasize the value of AI-driven approaches to aid developers understanding the larger implications of complex game mechanics. Similarly, Istamar et al. utilize reinforcement learning to assess game balance of an educational economy game [29], yet sticking to a single dimension again.

Up to this point, even when taking place in games with a long history of shifting balance, scientific work did not venture into understanding, visualizing or adjusting balance on a higher-dimensional scale. To a certain extent, this is tackled within the competition around adversarial Pokémon battles founded by Reis et al., each specified by its species, trained attributes, selected move set, and synergies to other Pokémon in the team [56, 57]. In multiple publications, they elaborate how AI can adjust some of the mentioned variables to increase the set of viable alternatives. Similarly, Hernandez et al. approach the balancing of multiplayer games (e.g. fighting games) at design time by manipulating internal unit variables towards a more even win rate [27]. In order to counter dominant (sets of) strategies, Kavanagh et al. iteratively alter the balance of a test-bed RPG in favor of less viable options [33, 34]. What these projects have in common is that they operate across multiple configurations of the state of balance (or metagame), which are getting adjusted after assessment and deriving operations towards increased balance. Yet, this internal, iterative dimension of progression (across balance states) is only one possible direction, whereas the complexity of modern games highly suggests a multitude of interlocking factors. Jeon et al. introduced an environment that simulates, calculates and automatically balances boss raid combat of MMORPGs set in a simplified test bed game [32]. They do generate skill configurations for players following multiple parameters (thus ending in a higher-dimensional space per skill), yet focus on algorithmic contributions for automated content generation, while our approach aims to understand and inform about balance in existing games. In our previous work, we modeled the combat behavior of a whole player community from a server of the MMORPG *Aion*, simulated their proficiency throughout permuted sets of encounters, and proposed approaches to improve the balance in between classes [48, 49]. While this would at least go into the direction of *encounter progression* (which we will cover later), it still leaves much directions open, and does not necessarily satisfy the understanding of balancing in the player community [50, 52].

In the remainder of this paper, we set out to overcome this gap of knowledge with the aid of comprehensive simulation, community expertise and interactive visualization.

2.2 Visualization

With regards to game analytics, visualization is one of the main paradigms of conveying information and context in an intuitive and explainable manner [20, 69]. Major advancements have been made to deliver insights for the game industry (e.g. data mining, classification or prediction) [16, 47], targeting measures against churn [26], facilitating content generation [58], or easing the burden of testing [3, 53], among other areas. Beyond straight-forward use cases, several approaches follow academic interests concerning

fundamental psychological or technical concepts [17, 24, 25, 72], often employing visualizations to gather insights for researchers or analysts [11]. Prominent implementations target spatio-temporal movement [2, 41, 42, 70, 71], decision making of individual or aggregate players [36, 39, 43, 66] or higher-level metrics and statistics [18, 65]. In addition to interests of industrials and academics, player communities have shown increasing levels of interest in inspecting, interpreting and learning from their own data as well, which leaves this angle vital to attend [45, 51, 52].

As soon as the dimensionality of the data of interest exceeds what is trivially observable by humans, visualization approaches have been struggling in intuitiveness and interpretation, which most often led to dimensional reduction. Recent research from Mane et al. introduce their analytical tool for 4D flight simulator telemetry (plane positions in 3D coordinates over time) and break these down into a more graspable 2D view of events of interest over time [40]. In the similarly recent *GAMR* tool of Sargolzaei et al., player traces from mixed reality games are visualized as trajectories through 3D space [59]. To reduce clutter when inspecting multiple traces at once, the authors flatten those down into traditional 2D heatmaps. As long as this dimensionality only concerns movement data (or events in space), the ease of visual interpretation can be maintained with an acceptable loss of information. To extend these approaches into an even higher-dimensional case though (here: multifaceted balance variables), we propose to foster understanding of these constellations by local slice visualizations, linked interactive plots, and supporting metrics that quantify the impact of the dimensions onto overall balance.

3 Approach

To address the complex problem of multifaceted factors, we developed a generic combat simulator that can produce expected damage values for every state-action pair in video games (and sequences thereof). The (participatory) design, and technical outline of this tool are explicated in related prior work [46], whereas the contribution of this paper lies in the advancement of methods towards multi-dimensional balance criteria. To make sure that this is relevant, complex enough and to reach a large target audience that is engaged in discussion, analysis and critique, we adapted this game simulator to the *Game of the Year 2023: Baldur’s Gate 3*. The simulator is capable of transforming the entire (highly probabilistic) combat logic of *Baldur’s Gate 3* into deterministic, expected values (which ensures reproducibility and comparability across benchmarks). We implemented all of the game’s 12 classes throughout their whole level progression (including hundreds of spells, actions and passive modifiers). After establishing this potent basis for computational exploration, we conducted an initial benchmark, providing a first assessment of balance between the 12 classes under fixed circumstances. Using the computed damage values of this, we inferred potentially drastic balance deviations and subsequently aimed at broadening our conception with the collective knowledge of its player community.

3.1 Baldur’s Gate 3

To give a short description about the game environment chosen for this work, *Baldur’s Gate 3* is a tactical, turn-based role-playing game

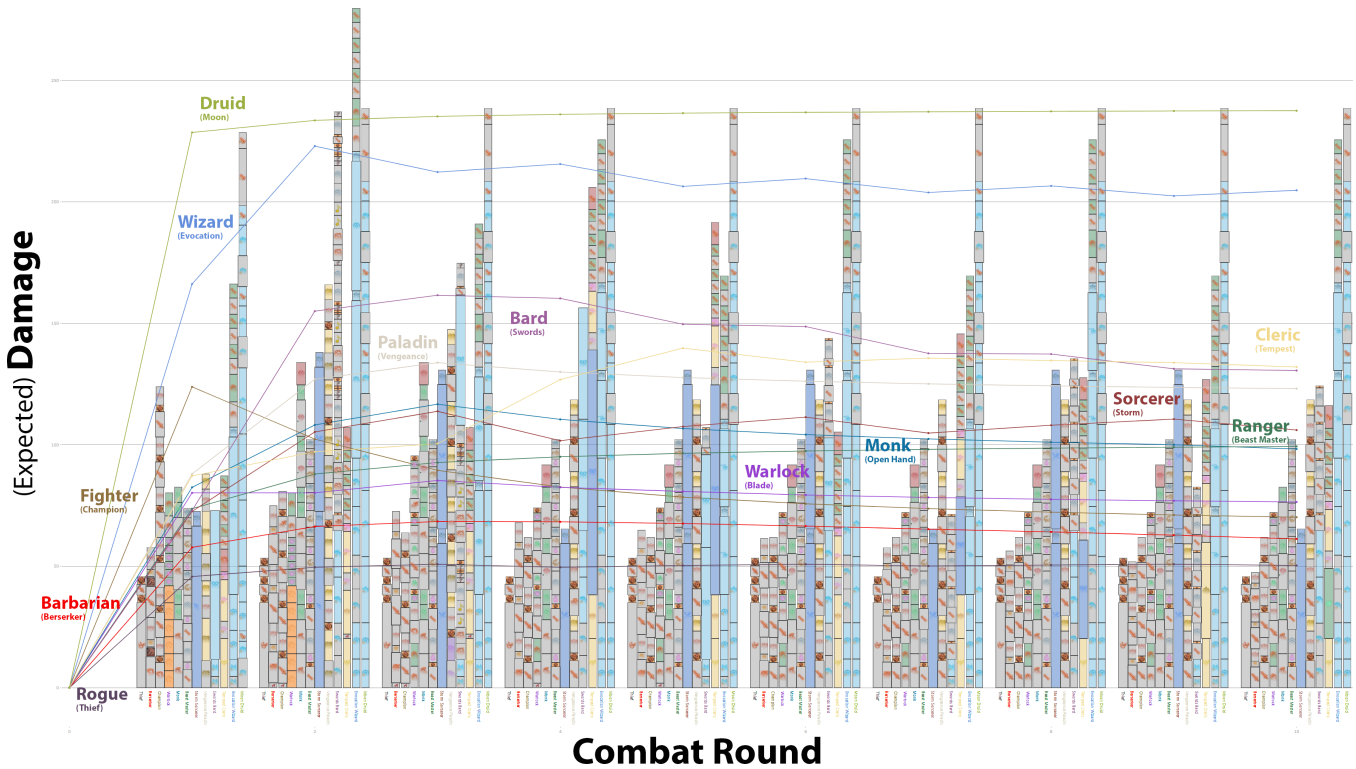


Figure 2: Progression chart showcasing the (*Classes × Combat Progression*) outcome of the initial benchmark. Levels of all 12 classes were fixed at 12, the length of the combat at 10 rounds (horizontal axis), and the enemy was set to a default target with average stats. Stacked bars indicate the skills (attacks or spells) that every class should ideally execute in order to perform maximal damage on the long term (where the bar height is stating the raw expected damage of that action). Line charts add the average damage per round value (vertical axis) per class over the course of the simulated combat, where dominating trends become apparent quickly (*Druid* outperforming all other classes in this constellation, while *Rogue* is unable to compete with any other). This Figure can be interactively viewed at <https://nevermindcreations.de/bg3optimizer?sessionID=benchmark01>

released by Larian Studios in 2023. It is built on the Pen and Paper ruleset of *Dungeons & Dragons: 5th Edition* and extends this by some custom skills, items and configurations. Players can engage in the campaign in solo mode or with up to three fellow players. Up to a party of four in total, companions can be recruited that enrich the game with narrative, quests, romance options and are controllable in combat just as the player character. There are 12 classes that any character can choose from (or combine), which unlock different attacks, spells, passive modifiers, triggers and other perks. Combat is executed by taking turns, in which round-specific resources such as Actions, Bonus Actions or Movement Speed can be spent to execute attacks or other moves. For caster characters, these resources can be combined with Spell Slots (or other resources) to cast a multitude of spells. Depending on the character attributes, the target(s) stats, their spatial configuration, passive feats, equipment, (dice-roll) probabilities and other factors, those actions can inflict damage, heal characters, apply beneficial or malicious statuses, change the environment or cause other effects. As there is a vast majority of decisions when it comes to equipment choices, build configurations, ways to tackle opponents or fights, item utilization, adaptability to randomness, and the strategy and order of

own planned actions, *Baldur's Gate 3* is a highly complex game that enables interesting exploration into intricate game mechanics, and could highly benefit from explainable and comprehensive balance assessment methodologies, rendering it a viable use case for this work's purposes.

3.2 Benchmark

Selecting an appropriate environment (and configuration) for the initial benchmark should not appear arbitrary, but ideally examine and give insights about why certain builds are stronger (or weaker) in certain situations than others. To encompass the full mechanical complexity of the classes, we thus defined the level of the benchmark to happen at the maximum obtainable level in the main game (12). It was reasonable to expect that different classes follow different curves of damage progression, wherefore we fixed the number of combat rounds to 10, as this would allow us to identify trends between classes with a short-term burst focus and classes that benefit from long-tailed sustained damage. As combat requires a target, we chose the way of the least resistance (and deviation) for the start, and configured a non-acting opponent with infinite hit points but no special strengths or weaknesses (more precisely:

10 Armor Class, ± 0 to all Saving Throws). Combat followed the “Tactician” rule set of the game, all classes had only access to comparable (start) equipment and had no influence from other players (or companions). The build for every class was optimized for the largest potential damage output, including their passive modifiers (feats), selected sub-classes and spells.

Given this configuration, the simulator used heuristics-driven reinforcement learning (explicated in our introductory paper of the tool [46]) to calculate the optimal execution of action sequences (rotations). This step was necessary to ensure that all classes are judged on the basis of their full potential, as action selection and order are highly important for the outgoing damage (Example: The spell *Chain Lightning* deals 10-80 damage to the target [on average: 45], but if the action *Destructive Wrath* is executed before, it will inflict full 80 damage instead. If the spell *Create Water* was cast one to three rounds before that, the enemy is even vulnerable to all *Lightning* damage, increasing this to 160 damage). Other mechanics have even longer dependencies or require complex setups. Through extensive testing and aligning outcomes with in-game results, we made sure that the simulator would always follow the most optimal path with respect to outgoing damage. Finally, when performing the computation of the best possible rotation for all 12 classes, the simulator would display the output in numerical values and visualizations as depicted in Figure 2.

While the outcomes of the initial benchmark paint a clear picture in favor for classes such as the *Druid* or *Wizard*, we also directly acknowledged that this can be heavily influenced by the choice of our just outlined benchmark configuration. To anticipate some of the outcomes already, *Druid* is less dependent on having powerful equipment than any other class, and *Druid*, *Wizard*, and *Bard* can generate vulnerability towards (parts of) their own damage, effectively doubling the damage from these skills. All of them have also options to summon elemental creatures that benefit from the benchmark conditions and have buffs that increase their own potential even further. While these points already indicate some of the flaws of this one-shot benchmark, we aimed to complete the full extent of factors that could influence this, and tailor the tool's capabilities to a promising target group, as outlined in the following section.

4 Study

To ensure that outcomes of the benchmark and the underlying methodology align with the general goals of the game, we reached out to its player community. We outlined the idea, presented the (*Classes × Combat Progression*) plot (see Figure 2), as well as similar plots across (*Level Progression × Combat Progression*) for every particular class, gave them access to the interactive tool⁴, and captured their general stance, sentiment, ideas for further directions, and requirements regarding the tool over the course of one week.

4.1 Procedure

After configuring and computing the formerly outlined benchmark, we recorded a (~45 minute-) video explaining the technical implementation, analyses and outcomes, and published this through the most important channels for the player community

(the *r/BaldursGate3* subreddit, YouTube, Larian Forums)⁵. Together with a written description of our approach, we invited them to test the tool on their own, so that they could configure own builds, constellations, and benchmarks, as we aimed to advance their understanding of the in-game mechanics interactively. Afterwards, we asked custom qualitative questions towards our research questions, as listed in the following.

4.2 Material

To get an impression of the amount of players we reached from the community (and thus, if this is representative), we recorded absolute number of views across the channels, the number of unique users that executed at least one simulation in our tool, and the number of unique participants that supplied answers to our custom qualitative questions, posed as follows:

(Regarding the benchmark):

- Did you expect this ranking? Which classes came out surprisingly strong (or weak)?
- Did you discover some strategy that you didn't know or consider before?

(Regarding directions of balance assessment):

- If you are interested in such analyses, what would you be next interested in?
- Which factors have not been considered so far, but are important for balance in your opinion?

(Regarding the tool):

- Would you like to use such a tool for your own Theorycrafting as well?
- If yes, what features are you missing?

4.3 Participants

At the time of wrapping up the study, we registered over 500,000 viewers of our benchmark and tool introduction when concerning all channels. 1,151 unique users conducted over 48,000 different simulations between different classes, within the same class (but different levels), or between different builds or strategies of a class. 102 unique respondents gave feedback and answers to the formerly outlined questions.

4.4 Qualitative Analysis

After collecting the community responses, we categorized them into the particular questions they could answer (cf. Section 4.2) and whether they contained (positive or negative) sentiment. Afterwards, we applied either binary classification directly (for binary-coded questions, e.g. *if they learned something new*) or thematic analysis [12] (for more open questions where we wanted to derive themes, e.g. *which factors are important for balance*). This thus ended in either percentage countings or a set of themes from which the most popular and important outcomes are listed and discussed in the following sections.

⁴<https://nevermindcreations.de/bg3optimizer>

⁵“Which Baldur's Gate 3 class does the most damage?” I benchmarked them all!: <https://www.youtube.com/watch?v=817O4EH9FGQ>

Table 1: List and description of Progression Dimensions, as derived from the study outcomes (*: already implemented priorly)

Dimension	Description
* Classes	The set of available/playable game element options to balance.
* Level Progression	The course of class evolution throughout the entire game.
Combat Progression	The duration of combat as can be encountered in the game, revealing balance differences between short-term and long-term classes.
Item Progression	The change in efficacy of a class when gradually having access to better items throughout the game (as some classes are more dependent on equipment or consumables than others).
Encounter Progression	The set of different combat settings throughout the game (including the amount of enemies, their difficulty, frequency, etc.).
Difficulty Progression	The impact of the game’s difficulty setting (e.g. Easy/Medium/Hard pre-sets, or online Un-ranked/Ranked matches) on the class.
Challenge Progression	The impact of a player’s individual skill on the outcome of the class efficacy (some classes benefit more than others if controlled by a player with better strategic skills or proficiency in reaction time/muscle memory/spatial awareness, etc.).
Composition Progression	The change in efficacy of a class when adding support from other parties (e.g. companions, fellow players).
Support Progression	The change in efficacy of <i>other</i> classes when adding support from this class.
Temporal (Meta)-Progression	The change across major milestones (such as game update patches), which often tilt the efficacy of popular classes in favor for classes that have been neglected for a longer time.
Beyond-Combat Utility	The added value of a class when exposed to a series of situations out of combat (which can highly diverge between classes).

5 Results

Regarding sentiment, the approached community reacted mostly positive towards our approach and the presented benchmark. 88% of replies with clear sentiments were positive (e.g. “thats a really good approach to the problem”), where the remaining 12% showed doubts that *any* benchmark would be representative to in-game combat (e.g. “calculations not based on reality are worthless.”). This generally positive trend is furthermore reflected in the high rate of Reddit upvotes (97%) and the YouTube like/dislike ratio (97.1%).

Only 17% of statements on the surprisingness of the results were pessimistic about any potential insights (e.g. “everything discovered from this ephemeral analysis was already well-known”), despite the majority of participants found new insights or surprising results (83%, e.g. “this video made me realize I have so much more to learn about this game”, “that was an in depth insight”).

Regarding further factors or dimensions that can affect balance, participants mentioned plenty of possible directions. First of all, the length of combat was fixed for comparability in this benchmark, yet a game most often consists of fights of different lengths. Likewise, there are certain difficulty differences between fights, but also classes. As we only looked at combat performance here, multiple subjects voiced that out-of-combat utility can make a class viable (such as Rogues’ ability to steal, or Charisma-based classes’ capability of succeed in dialog checks). Another important point is the progression of items (in this case, we combine mentions of equipment and consumable factors), as different classes can scale differently strong with the access to stronger items. The presence or absence of companions also makes a big difference on someone’s viability, as some classes might be more self-sufficient, but others could positively escalate with the proper support. Eventually, the factor of risk was communicated, as some builds might be more

prone to adversities in combat than other, more robust classes. Beyond these, there were some very game-specific remarks, such as the ability to pre-cast combat actions before engaging, the level of control classes can output (which however was reflected in the expected damage computation of the benchmark), or the utilization of meta-class abilities (in this specific case, Illithid powers). Importantly, that also entailed the possibility of Multiclassing in BG3, which most often produces stronger builds, but we refrained from benchmarking these in the initial setup to cleanly attribute outcomes to the particular classes.

No particular demands for tool features were requested, but only the openness to extend the database, which carried out a special importance to games like *Baldur’s Gate 3* with (already) a lot of added build options, levels, and content through modding.

6 Discussion

The general well-received impression and ability to create tangible insights about game concepts, dynamics and overarching balance left an optimistic view about this direction of tool-assisted balance analysis. From the breadth of players that viewed and used our approach (and a good range of different views and statements), we assume that we reached a representative sample of the community. While some participants voiced a lack of coverage about the big picture (with regards to the full complexity of the target game), this is most likely attributable to the fact that the initial benchmark did not cover all dimensions necessary for the players’ conception of balance – which to comprehend was however our main motivation of this work. Based on the responses towards balance aspects not covered so far, we compiled the following dimensions, added to the already implemented dimensions for completeness (see Table 1).

Table 2: Configuration of the initial benchmark (cf. Section 3.2) versus this of the multi-dimensional progression benchmark, classified by the progression dimensions of Table 1

Dimension	Initial Benchmark	Progression Benchmark
* Classes	[Barbarian, Bard, Cleric, Druid, Fighter, Monk, Paladin, Ranger, Rogue, Sorcerer, Warlock, Wizard]	[Barbarian, Bard, Cleric, Druid, Fighter, Monk, Paladin, Ranger, Rogue, Sorcerer, Warlock, Wizard]
* Level Progression	[12]	[1,2,3,4,5,6,7,8,9,10,11,12]
Combat Progression	[10] (Rounds)	[1,2,3,4,5,6,7,8,9,10] (Rounds)
Encounter Progression	[Default]	[Default, Commander Zhalk, Grym, Apostle of Myrkul, Ansur, The Netherbrain]
Composition Progression	[Solo]	[Solo, Buffed (Haste), Enemy Debuffed (Perilous Stakes), Buffed & Enemy Debuffed]
Dimensionality	1	5
# Conditions	12	34, 560

For the sake of generalizability, the dimension of *Classes* does not necessarily have to refer to in-game classes in a strict sense. It is very well applicable to concepts of other games that use specific characters instead of classes (like heroes in *Overwatch* [10], champions in *League of Legends* [22] or even weapons in class-less games like *Counter Strike 2* [67], if they are comparably available and largely determine gameplay). This metric can also be treated more specifically, like incorporating all possible subclasses of a class (in games that feature these), or even all possible build configurations of a class (including e.g. multi-classing), without changing the semantics of this dimension. This list does not claim to be exhaustive, and not all of these are applicable to all kinds of games and genres. Especially by approaching communities and developers beyond a single game, we are looking forward to complement dimensions to this framework in future work. Nevertheless, we aim to answer our **RQ1** by providing a series of player community-defined dimensions that are closely related to the perception of balance in Table 1, together with short explanations.

While the introduction of these additional dimensions could increase the informational breadth about balance, it also complicates the analysis and interpretation of outcomes, since high-dimensional spaces are drastically harder to interpret. Therefore, we conclude the paper with a continuation of the original benchmark into the multi-dimensional balance space and offer insights through slices of visualization, as well as guiding metrics.

6.1 Multi-Dimensional Progression Balance Benchmark

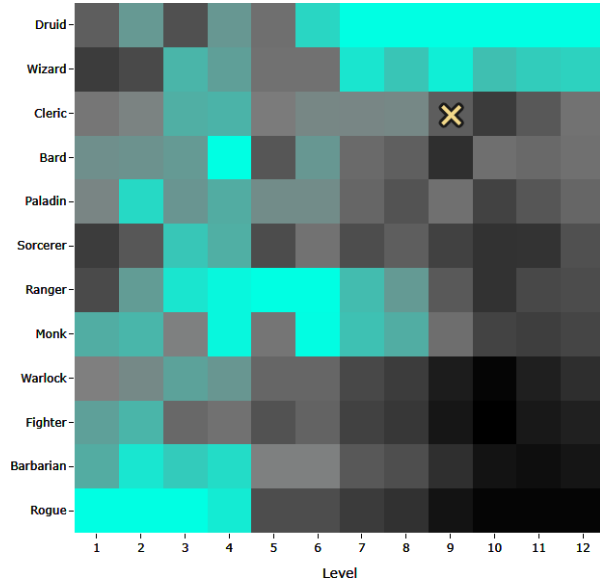
To extend our initial benchmark with the aggregated dimensions of the community survey, we reform the first setup into a higher-dimensional **progression** benchmark (both described in Table 2). For the sake of considering the computational cost, we focus on the (in our opinion) most informative progression factors, and only a subset of interest (e.g., six significant encounters at key positions in the game instead of likely hundreds of encounterable combat settings). We first executed the permutation of all possible combinations of the extended dimensions, leading to 34, 560 different cases and performance results. This could be averaged into mean outcome scores (per class) that would indicate a compressed aggregation of

the overall performance of that class, but this would leave us with a dead end in understanding why (or, in which contexts) which class over- or underperforms. As **RQ2** aims to explicitly unveil explanations with respect to balance dimensions, we propose further steps to investigate these, as covered in the following section: explanation by exploration and explanation by aggregation.

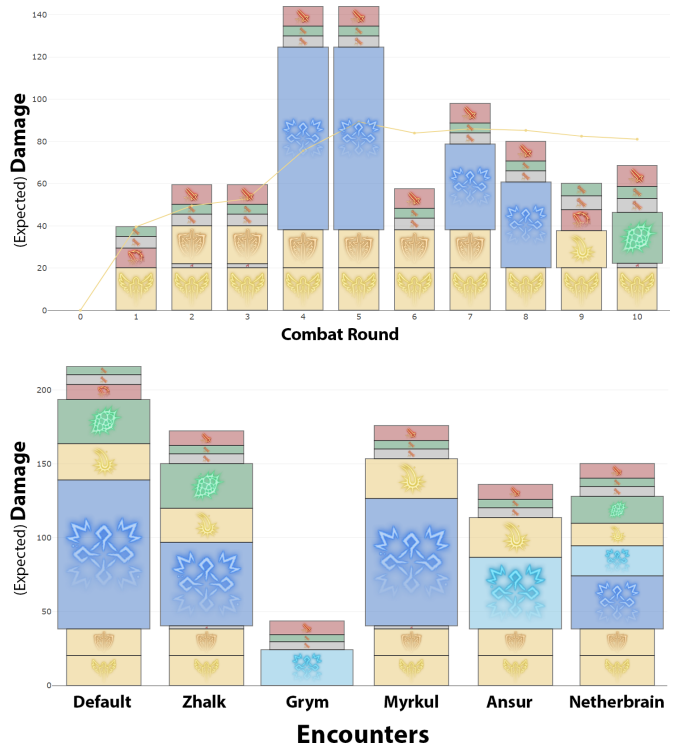
To prepare the high-dimensional data for more intuitive visualizations (exploration), we calculate slices of the results by flattening the values of all dimensions (except for two) into an average value that represents a single cell in the remaining two-dimensional matrix (which can be commonly visualized in a heatmap). For the aggregation part, we calculated a series of global statistics per game element (class), which could yield information for particular insights.

6.2 Outcome Measures

Even if we compute the combinatorial explosion of the just expanded benchmark, this does not necessarily entail that it would yield tangible insights already. To infer these, we refer to (complementary) pathways of reasoning, both implemented in the tool of this work: visual analytics and aggregate metrics. Since higher-dimensional outcomes are hard to illustrate and grasp from a glance, we propose to extract and visualize two-dimensional slices (see Figure 3a) that aggregate all other dimensions into cells of average color values, and present these in interactive context. That means that, in our published tool, these can be actively explored by hovering over a particular heatmap position, which then reveals information about the values of the other dimensions in additional plots displayed in parallel (see Figure 3b). Judging from the subjective impression of Figure 3a, no class appears to be utterly dominated across the field, as even classes that appeared as the weakest in the initial benchmark (e.g. *Rogue*, *Barbarian*) shine in their *niche* of the lower levels (making them excel in early game), and classes that appeared to be average (e.g. *Ranger*, *Monk*) actually outperform all remaining choices in the midgame. Conversely, classes that are very strong late-game (e.g. *Druid*, *Wizard*) have a harder time to compete in the first half of the level progression. With the special concept of *multi-classing* in *Baldur’s Gate 3*, this can also inform valid cut-off levels after which professions are no longer adding more than their alternatives.



(a) Heatmap visualizing the slice of (*Level Progression × Classes*). Cyan values correspond to the (relatively) highest (damage) scores in this slice, whereas black cells indicate the lowest outcomes.



(b) Top: (*Average DPR × Combat Progression*) slice for the selected heatmap parameters (*Cleric / Level 9*). Stacked bars indicate the average expected damage per skill (referenced by their in-game icons), if following the optimal skill rotation.

Bottom: (*Average Damage × Encounter Progression*) slice for the same selection. As different opponents can have different resistances, the optimal damage distribution can deviate from the default (e.g.: *Grym* and *Ansur* are immune to **Lightning** damage (and others), whereas the *Cleric* has to resort to its second best option, **Cold** damage. The *Netherbrain* becomes immune to certain damage times temporarily, hence optimal rotations alternate between damage types.

Figure 3: By breaking down the number of progression dimensions to two, the slice heatmap (a) can communicate the impact of two factors of interest immediately (while the other dimensions are aggregated, and their average value determines the color shade). To not lose information about the other dimensions, users can navigate the heatmap (as indicated by the yellow ×), which displays the remaining dimensions in parallel plots accordingly (b). This Figure can be interactively viewed at <https://nevermindcreations.de/bg3PB>.

While this could empower developers, analysts and players to explore concrete outcomes and relations to other dimensions already, these might be still subject to manual interpretation, and are unlikely to be exhaustively explored. Thus, we also communicate the following metrics as condensed output criteria of multi-dimensional balancing (see Table 3).

Regarding **RQ2**, we are contributing to the question which methods can advance the understanding of game balance variables with the following arguments:

- The majority of the surveyed community of this case study's game indicated that they learned new insights through our interactive tool (83%).

- Adding multi-dimensional slice visualizations that are linked to slices of adjacent dimensions allows to draw conclusions that were unavailable with traditional, one-dimensional balancing measures (cf. Figure 3).
- Aggregate outcome metrics condense and can yield information about abstract qualities of the game or its elements (cf. Table 3).

We contribute our holistic approach (combining the game-agnostic simulator, the visual analytical tools, and the automatically calculated outcome metrics as an open source framework, so that it can be applied to any game beyond *Baldur's Gate 3* (or the structurally very different *Guild Wars 2* and *Borderlands 3* of our related work) [46]).

Table 3: Outcome metrics of multi-dimensional progression balance assessments.

Metric	Description
<i>Impact</i>	The higher the <i>impact</i> of a dimension onto a class is, the more important it is for this class (relatively to other classes). Examples: - The <i>impact</i> for <i>Level</i> on <i>Druid</i> is higher than for any other class, since averaged over all levels, <i>Druid</i> benefits most from leveling up. - The <i>impact</i> for <i>Encounters</i> on <i>Monk</i> is lower than for all other classes, since its damage ignores most resistances, making it a versatile option for any opponent.
<i>Progression-Δ</i>	The higher the <i>progression-Δ</i> of a dimension for a class is, the higher the variance within the dimension for that class. Examples: - The <i>progression-Δ</i> for <i>Composition</i> on <i>Sorcerer</i> is lower than for all other classes, since the <i>Sorcerer</i> benefits the least from support of others (it already can provide the most important buff and debuff on its own). - The <i>progression-Δ</i> for <i>Level</i> on <i>Rogue</i> is lower than for all other classes (since it has a comparably uniform level progression).
<i>Dominance</i>	The <i>dominance</i> of a class is equal to the number of other classes it outperforms throughout all conditions (can be queried for a particular dimension, or a particular value of a dimension). Examples: - The <i>Rogue</i> dominates more classes than any other class when <i>Level</i> is 1. - The <i>Rogue</i> dominates less classes than any other class (overall).
<i>Divergence</i>	The deviation of the <i>dominance</i> of a class from the “ <i>expected ideal dominance</i> ” (0.5, or “dominating half of the other classes on average”) across all dimensions. Classes with low <i>divergence</i> are close to dominating the same amount of classes in any condition, while classes with high <i>divergence</i> are exceptionally good in some cases and exceptionally bad in others.
<i>Balance</i>	The average value of <i>divergence</i> for all classes. Could be used to compare two different game configurations (or benchmarks).
<i>Diversity</i>	The variance of <i>divergence</i> over all classes. A game (configuration) could achieve ideal <i>balance</i> (0.5) by just offering perfectly symmetric options. This however would leave the <i>diversity</i> score at 0, whereas higher values here go along with more extreme distributions of class strengths/weaknesses across all dimensions.

6.3 Implications

As we can draw from previous literature, a class can be considered viable if it allows a player to win the game [64]. However, this can already be reached by either a very low difficulty level (in single-player games) or by perfect symmetry (in single- or multi-player games). Both options would however be arguably subpar for player experience, as there would be no (game-mechanical) appeal to any option, which eventually renders decisions meaningless and is to be avoided [63]. Beyond balance, our approach also measures divergence, which can be translated to how niche a class can be. If a class is high in divergence, but low in balance, then a game might offer valuable niches for this class to shine, but too few of them to make this class a meaningful pick (or it might be disproportionately strong, as would be indicated by its relative dominance). If a class is low in divergence, but high in balance, it might be a safe-to-play class that however does not yield any surprising moments during the course of the game. If it instead is high in both divergence and balance, it should conveniently align itself into play, in some situations being satisfyingly good, while acceptably bad in others.

On a higher level, a game can be balanced without being necessarily diverse. Only if many opportunities exist in which different classes can shine (rendering decisions between them meaningful), and only if these are equally available for the range of class options, we can consider a game balanced and diverse. Eventually, these decisions are in the hands of game designers and developers, but with this work, we are aiming to contribute towards this high goal

by providing tools and measures to improve informed decision making.

7 Limitations & Future Work

The underlying foundational metric for this approach and benchmarking was damage. As a single-player game, it could be argued that our case of *Baldur’s Gate 3* might not necessitate perfectly balanced damage output - but that picking the class of preference is a much more important appeal for a good amount of players (irrespective of the efficacy). Still, we argue that maintaining that every of these classes stays viable is a core element of balance for any game, as effortlessly doing too much damage would render the combat aspect of a game as too easy, usually leading to boredom. On the other hand, frustration caused by the inability to reach damage potentials necessary to overcome the game’s challenges could quickly lead to frustration, equally harming flow. In the worst case, imbalances could discourage players to play their preferred class, flair or theme, which would greatly limit player experience. With special reference to this use case, the introduction of the honour mode and mod support in the latest patches drastically increased the challenge of the game, the interest in mechanical insights in classes, and even the need to play well-rounded builds (according to the participants of this work).

Certainly, damage output is also not the only parameter that is decisive for the balance between game elements (classes). To a certain degree, we already measure and indicate the impact a class

has on fellow players/companions by the dimension of support progression (see Table 1). Yet, this also only concerns supplementing offensive values, whereas dimensions like lowered risk or increased survivability should usually compensate for lack of offensive capabilities.

A proper evaluation about the usability and explainability of the higher-dimensional progression balance output viewer (and the accompanying metrics) is still necessary to draw valid conclusions about the eventual outcome of the final approach. We are following up with this assessment in upcoming work, but wanted to focus on the aggregation of progression dimensions in this paper. This state of our work might have also led to the lack of novel answers regarding **RQ3** (new features necessary to advance visual tools for game balance). To include different angles of users that might require explicit functionalities, this follow-up evaluation includes developers of the original game on top of the community surveyed here.

It is to be noted that increasing dimensionality can become expensive, especially with complex games and if combat outcomes can not be efficiently computed. From our initial 12 iterations, we quickly reached 34,560 conditions in the subsequent 5-dimensional progression benchmark. In theory, it is in the operator’s responsibility to configure meaningful, expressive and slim dimensions and values. In practice, this can be guided by following an iterative approach to increasing the benchmark magnitude: First, adding a single dimension at a time, and then using *progression- Δ s* on the value-level of a dimension, to identify considerable gaps in which conditions are too similar. The value range can then be shrunk to focus only on the most important milestones per dimension. In already published games, single outcomes could also be complemented or approximated by real statistics from recorded fights of the player base.

While we already covered a part of progression dimensions within *Baldur’s Gate 3*, we omitted the crucial player choices of complex equipment configuration and multi-classing, since we wanted to avoid an even larger combinatorial explosion for the launch of this tool. In immediate following work, we are already collecting the most popular, as well as the most promising builds that the *Baldur’s Gate 3* community came up with and shared through multiple channels. These builds, with all of their particular items, multi-class combinations and other build choices, will serve for a next, advanced benchmark that extends our initial insights.

Beyond *Baldur’s Gate 3*, this framework is already being applied to other games, with which we aim to improve the argument of generalizability. The proposed method is not constrained to a single genre, but generally applicable to games with choosable elements, i.e. those for which balance is an important criterion. In concurrent related work, we already applied this methodology to games of other genres (MMORPG, FPS) [46], but it is likely to be applicable to further genres as well (such as Fighting Games or RTS), as long as balance dimensions of interest can be pinpointed and performance metrics measured. These do not necessarily have to be tailored to combat, as e.g. cars in Racing Games also ought to be balanced, or ideally, there is a time and place to shine for those (e.g., cost/performance trade-offs, early vs. late game progression, or tracks that favor acceleration vs. speed).

As a long-term goal, iterative simulations and informative metrics could be key to unraveling automated game adjustment (towards both game balance and diversity). We are actively pursuing this direction of research in future work, and will evaluate its potential impact on the player community by publishing mods that incorporate these lines of balance decisions.

8 Conclusion

Ensuring balance between the choosable options a game has to offer is one of the keys to a well-rounded player experience and satisfying, meaningful decisions. While several scientific approaches already tackle the measurement and adjustment of balance, they are most often confined to a single dimension of comparison, e.g. just the set of classes to balance itself, while all other assumptions are fixed. However, a game does not only happen during a specific time frame, in a single configuration, on a single encounter, or in the end-game phase alone. Thus, we tackled the assessment of balancing variables at large by developing a precise simulator embedding game-playing AI and analytics to contrast all possible cases that could occur. We made this tool publicly available, demonstrated its functionality in an initial benchmark and invited the *Baldur’s Gate 3* community to use it, advance their understanding of in-game mechanics, and report on factors that are missing to complete their concept of what makes classes balanced. After interpreting their understanding, we extracted nine further dimensions of progression that affect the balance of a class across time, level, party size and more. At large, the major contribution of this work is to emphasize that games can be considered balanced if they realize a well-rounded composition of “*times and places to shine*” for all their game elements - wherefore we supply appropriate analytical tools and metrics.

Acknowledgments

We thank all participants for sharing their feedback and understanding of the game, its balance, and our approach to that. Even more, we are grateful to Larian Studios for developing *Baldur’s Gate 3*.

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Received 12 September 2024; revised 10 December 2025; accepted 16 January 2025